Statistics and Medicine

In Medical sciences, statistical tools are widely used. In order to test the efficiency of a new drug or to compare the efficiency of two drugs or two medicines, t - test for the two samples is used. More and more applications of statistics are at present used in clinical investigation.

Statistics and Modern applications

Recent developments in the fields of computer and information technology have enabled statistics to integrate their models and thus make statistics a part of decision making procedures of many organisations. There are many software packages available for solving simulation problems.

12.7 Limitations of statistics

Statistics with all its wide application in every sphere of human activity has its own limitations. Some of them are given below.

1. Statistics is not suitable to the study of qualitative phenomenon: Since statistics is basically a science and deals with a set of numerical data. It is applicable to the study of quantitative measurements. As a matter of fact, qualitative aspects like empowerment, leadership, honesty, poverty, intelligence etc., cannot be expressed numerically and statistical analysis cannot be directly applied on these qualitative phenomena.

2. Statistical laws are not exact: It is well known that mathematical and physical sciences are exact. But statistical laws are not exact and statistical laws are only approximations. Statistical conclusions are not universally true. They are true only on an average.

3. Statistics table may be misused: Statistics must be used only by experts; otherwise, statistical methods are the most dangerous tools on the hands of the inexpert. The use of statistical tools by the inexperienced and untrained persons might lead to wrong conclusions.

4. Statistics is only one of the methods of studying a problem: Statistical method does not provide complete solution of the problems because problems are to be studied taking the background of the countries culture, philosophy, religion etc., into consideration. Thus the statistical study should be supplemented by other evidences.

12.8 Types of Statistics

There are two major types of statistics named as Descriptive Statistics and Inferential Statistics.

Descriptive Statistics

The branch of statistics devoted to the summarization and description of data is called Descriptive Statistics

Inferential Statistics

The branch of statistics concerned with using sample data to make an inference about a population of data is called Inferential Statistics.

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